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1)

a) Find gradient $\nabla f(x, y)$

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

$$f(x, y) = (6x - x^2)(4y - y^2) \rightarrow g(x)h(y)$$

$$g(x) = 6x - x^2$$

$$h(y) = 4y - y^2$$

$$\frac{\partial f}{\partial x} = (6 - 2x)(4y - y^2) \quad \left(\begin{array}{l} \therefore \text{differentiating with respect to } x \\ \text{keeping } y \text{ term as constant} \end{array} \right)$$

$$\frac{\partial f}{\partial y} = (6x - x^2)(4 - 2y) \quad \left(\begin{array}{l} \therefore \text{differentiating with respect to } y \\ \text{keeping } x \text{ term as constant} \end{array} \right)$$

$$\therefore \nabla f(x, y) = \begin{pmatrix} (6 - 2x)(4y - y^2) \\ (4 - 2y)(6x - x^2) \end{pmatrix}$$

b) Hessian Matrix

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

i) $\frac{\partial^2 f}{\partial x^2}$

Differentiate $\frac{\partial f}{\partial x}$ with respect to x .

$$\therefore \frac{\partial^2 f}{\partial x^2} = \text{Differentiation of } \frac{\partial f}{\partial x} = (6 - 2x)(4y - y^2) \text{ with respect to } x.$$

$$\therefore \frac{\partial^2 f}{\partial x^2} = -2(4y - y^2)$$

ii) $\frac{\partial^2 f}{\partial y^2}$

~~Differentiate $\frac{\partial f}{\partial y}$ with respect to y~~

Differentiate $\frac{\partial f}{\partial y} = (6x - x^2)(4 - 2y)$ with respect to y .

$$\frac{\partial^2 f}{\partial y^2} = -2(6x - x^2)$$

$$\text{ii) } \frac{\partial^2 f}{\partial x \partial y}$$

Differentiate $\frac{\partial f}{\partial x}$ w.r.t $y = (6-2x)(4y-y^2)$ with respect to y .

$$\frac{\partial^2 f}{\partial x \partial y} = (6-2x)(4-2y)$$

$$\therefore H(x, y) = \begin{bmatrix} -2(4y-y^2) & (6-2x)(4-2y) \\ (6-2x)(4-2y) & -2(6x-x^2) \end{bmatrix}$$

c) Locate & classify Stationary Points:

Critical points

$$\nabla f(x, y) = 0$$

$$(6-2x)(4y-y^2) = 0 \quad \text{--- (1)}$$

$$(6x-x^2)(4-2y) = 0 \quad \text{--- (2)}$$

Solve (1)

$$6-2x = 0 \Rightarrow \boxed{x=3}$$

$$4y-y^2 = 0 \Rightarrow \boxed{y=4} \text{ \& \& } \boxed{y=0}$$

Sub x & y in (2)

$$x=3 \text{ in (2)} \Rightarrow (18-9)(4-2y) = 0 \Rightarrow 4-2y=0 \Rightarrow \boxed{y=2}$$

$$\therefore \text{Critical point is } \boxed{(3, 2)}$$

Now sub $y=4$ in (2)

$$(6x - x^2)(4-8) = 0$$

$$6x = x^2$$

$$\boxed{x=6}$$

$\therefore$ critical point is $(6,4)$ & $(0,4)$

Now sub $y=0$ in (2)

$$(6x - x^2)(4) = 0$$

$\therefore$ critical points are $(0,0)$ & $(6,0)$

$\therefore$ critical points are $(3,2)$ $(0,0)$ $(6,0)$ $(0,4)$ & $(6,4)$

Now substitute all the critical points in $H(x,y)$ to find local maxima if $\text{Det}(H) > 0$

$$\underline{H(3,2):}$$

$$H(3,2) = \begin{bmatrix} -2(4(2) - 2^2) & (6-2(3))(4-2(2)) \\ (6-2(3))(4-2(2)) & -2(6(3) - 3^2) \end{bmatrix}$$

$$= \begin{bmatrix} -2(4) & 0(0) \\ 0(0) & -2(9) \end{bmatrix} = \begin{bmatrix} -8 & 0 \\ 0 & -18 \end{bmatrix}$$

$$H(3,2) = \begin{bmatrix} -8 & 0 \\ 0 & -18 \end{bmatrix}$$

$$\det(H) = (-8)(-18) - (0)(0) = 144 > 0.$$

$\therefore$ Local Maximum is at point $(3,2)$

$$H(0,0) = \begin{bmatrix} -2(4(0) - 0) & (6 - 2(0))(4 - 2(0)) \\ (6 - 2(0))(4 - 2(0)) & -2(6(0) - 0^2) \end{bmatrix}$$

$$= \begin{bmatrix} -2(0) & (6)(4) \\ (6)(4) & -2(0) \end{bmatrix} = \begin{bmatrix} 0 & 24 \\ 24 & 0 \end{bmatrix}$$

$$\det(H) = (0)(0) - (24)(24) = -576 < 0.$$

$\therefore (0,0)$ is a saddle point

$$H(6,0) = \begin{bmatrix} -2(4(0) - 0^2) & (6 - 2(6))(4 - 2(0)) \\ (6 - 2(6))(4 - 2(0)) & -2(6(6) - 6^2) \end{bmatrix}$$

$$= \begin{bmatrix} -2(0-0) & (-6)(4) \\ (-6)(4) & -2(0) \end{bmatrix} = \begin{bmatrix} 0 & -24 \\ -24 & 0 \end{bmatrix}$$

$$\det(H) = (0)(0) - (24)(-24) = -576 < 0$$

$\therefore (6,0)$ is a saddle point

~~$$H(0,4) = \begin{bmatrix} -2(4/4) - 16 & (6-2(0))(4-2(4)) \\ (6-2(0))(4-2(4)) & -2(6/0) - 0^2 \end{bmatrix}$$~~

$$H(0,4) = \begin{bmatrix} -2(4/4) - 16 & (6-2(0))(4-2(4)) \\ (6-2(0))(4-2(4)) & -2(6/0) - 0^2 \end{bmatrix}$$

$$= \begin{bmatrix} -2(0) & 6(-4) \\ 6(-4) & -2(0) \end{bmatrix} = \begin{bmatrix} 0 & -24 \\ -24 & 0 \end{bmatrix}$$

$$\text{Det}(H) = 0(0) - (-24)(-24) = -576 < 0$$

$\therefore (0,4)$ is a saddle point

$$H(6,4) = \begin{bmatrix} -2(4/4) - 16 & (6-2(6))(4-2(4)) \\ (6-2(6))(4-2(4)) & -2(6/6) - 36 \end{bmatrix}$$

$$= \begin{bmatrix} -2(0) & -6(-4) \\ -6(-4) & -2(0) \end{bmatrix}$$
~~$$= \begin{bmatrix} 0 & 24 \\ 24 & 0 \end{bmatrix}$$~~

$$\therefore \text{Det}(H) = \begin{bmatrix} 0 & 0 \end{bmatrix} - (24)(24) = -576 < 0$$

$\therefore (6,4)$ is a saddle point

$$e) D = \{ (x, y) : x+y \geq 4, x \leq 4, y \leq 5 \}.$$

$$x=4 \text{ (vertical)}$$

$$y=5 \text{ (horizontal)}$$

$$x+y=4 \text{ (diagonal)}$$

$$y = -x + 4 \quad \text{slope} = -1.$$

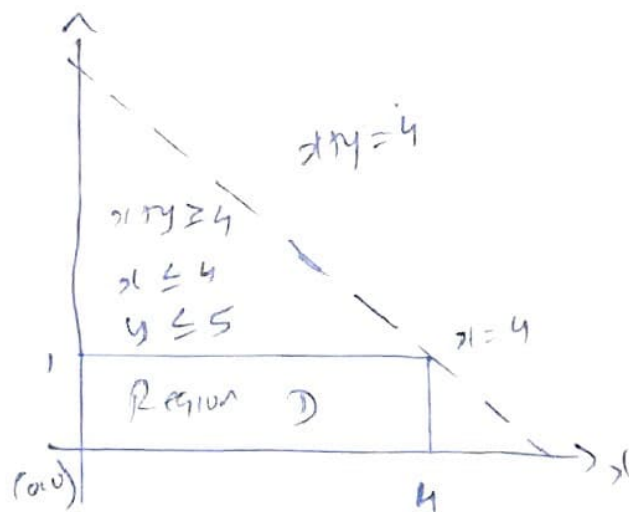
$$-x + 4 = 5$$

$$\boxed{x = -1}$$

$$y = -x + 4 \quad \text{at } x=4.$$

$$y=0$$

$\therefore$ Point of intersection is $(-1, 5)$ $(4, 5)$
 $(4, 0)$



e) Global extrema:

$$f(3,2) = (6 \cdot 3 - 9)(4 \cdot 2 - 4) = 9 \cdot 4 = 36.$$

Interior critical point = 36.

Edge $y = 5$ ($-1 \leq x \leq 4$)

$$f(x,5) = -5(6x - x^2) = 5x^2 - 30x$$

$$\text{Minimum at } x = 3 : f = 5(9) - 30(3) = -45$$

$$\text{Maximum at } x = -1 : f = 35$$

Edge $x = 4$ ($0 \leq y \leq 5$)

$$f(4,y) = 8(4y - y^2)$$

$$\text{Maximum at } y = 2 : f = 32$$

$$\text{Minimum at } y = 5 : f = -40$$

Edge $x+y=4$ ($-1 \leq x \leq 4$)

Parameter $x = t, y = 4 - t$

$$f(t, 4-t) = t^2(6-t)(4-t)$$

$$\text{Maximum at } t = -1 : f = 35$$

$$\text{Minimum at } t = 0 \text{ or } t = 4 : f = 0$$

$$\text{Global maximum } f_{\max} = 36 \text{ at } (x,y) = (3,2)$$

$$\text{Global minimum } f_{\min} = -45 \text{ at } (x,y) = (3,5)$$

2)

a) There are 5 working days in a week and there are 4 appointments.

Each 4 appointment can be distinctly scheduled in the 5 days in 5^4 ways.

$$5^4 = 5 \times 5 \times 5 \times 5 = 25 \times 25 = 625$$

-X.

Note that this is valid only when ~~no~~ repetitions are allowed -X.

$\therefore$ 4 appointments can be arranged / appointed / scheduled in 625 ways for the 5 working days.

b) We need to count the number of ways 4 distinct appointments can be scheduled between 5 days such that no day gets more than 2 appointments & more than 1 appointment per day.

i) More than one appointment $\Rightarrow$ $\overset{\text{M T W T F}}{(1, 1, 1, 1, 0)}$

This can be done by appointing 4 appointments over the 4 selected days $\Rightarrow 4! = 24$.

There are 5 ways to leave out the one day. $\therefore 5 \times 24 = 120$

$\therefore$ There are 120 ways 4 appointments can be scheduled over the 5 days with not more than 1 appointment per day.

ii)
Now we need to calculate how many ways 4 appointments can be scheduled over 5 days with no more than 2 appointments per day.

This can be done in two ways.

Type 1: (2, 2, 0, 0, 0)

Choose 2 days to get 2 appointments

$\Rightarrow 10$ ways

ways to divide 4 clients to two unordered pairs $= \frac{1}{2!} \binom{4}{2} \binom{2}{2} = 3$.

Now assign each pair to a chosen day pair

$\Rightarrow 10 \times 2! = 20$.

$\therefore$ Total ways = $3 \times 20 = 60$ ways

Type 2: ~~(2,1,1,0,0)~~

(2,1,1,0,0)

Choose 3 days out of 5.

$$\therefore {}^5C_3 = 10.$$

Choose one of the day to receive two appointments = 3.

Now choose 2 clients to go to the remaining two days = $4!2 = 6$.

The remaining 2 clients can go to two singleton days in $2!$ ways.

$$\therefore 10 \times 3 \times 6 \times 2 = 360.$$

~~Type 3:~~ Type 3: (1,1,1,1,0) which is already done in (i).

$\therefore$ Total valid arrangements with no more than 2 per day is

$$60 + 360 + 120 = 540$$

c)

i) At least one appointment on Monday:

$$\text{Total} \Rightarrow 5^4 = 625$$

No. of arrangement with no appointment on Monday $\Rightarrow 4^4 = 256$.

$$\therefore 625 - 256 = 369.$$

$\therefore$ There are 369 ways appointments can be scheduled with at least one appointment on Monday.

ii) Appointments on both Monday + Tuesday.

$$\text{Total} = 5^4 = 625$$

$$\text{without Monday} = 4^4 = 256$$

$$\text{without Tuesday} = 4^4 = 256$$

$$\text{without Both} = 3^4 = 81$$

$$\therefore 2) 625 - 256 - 256 + 81 = 194$$

$\therefore$ There are 194 ways appointments can be scheduled on both Monday + Tuesday.

d) Exactly 3 days have appointments.

We must select 3 distinct days such that all appointments are placed on them & no day is empty.

$\therefore (2, 1, 1)$: One day gets 2 appointments others get 1 each

$\therefore$ Choose 3 days $\Rightarrow 5(3) = 10$.

Choose which gets 2 appointments = 3.

Choose 2 of the 4 clients to go to that day $\Rightarrow 4(2) = 6$.

Remaining clients go to the remaining two days $\Rightarrow 2! = 2$.

$$\therefore 10 \times 3 \times 6 \times 2 = 360.$$

$\therefore$ There are 360 ways of scheduling 4 appointments with exactly 3 days.

e) x_1, x_2, x_3, x_4, x_5 are different workload profiles.

$$x_1 + x_2 + x_3 + x_4 + x_5 = 4 \quad x_i \geq 0$$

$$\Rightarrow {}^{4+5-1}C_4 = {}^8C_4 = 70.$$

f) How many workload profiles where all $x_i \leq 2$.

$$\underline{(2, 2, 0, 0, 0)}$$

$$\Rightarrow {}^5C_2 = 10.$$

$$(2, 1, 1, 0, 0)$$

$$\Rightarrow \frac{5!}{1! 2! 2!} = 30.$$

$$(1, 1, 1, 1, 1)$$

$$\Rightarrow {}^5C_1 = 5.$$

$\therefore$ Total = $10 + 30 + 5 = 45$ workload profiles.

3) Let A : first, second + fourth are Tails
 Z : 6 flip sequence reads palindrome

a) Are A and Z independent?

Event A

Only 3, 5, 6 are free $\Rightarrow 2^3 = 8$ outcomes

$$P(A) = \frac{8}{64} = \frac{1}{8}$$

Event Z

first 3 are free. Remaining 3 are reversed/forced.

$$\Rightarrow 8 \text{ outcomes} \Rightarrow \frac{8}{64} = \frac{1}{8}$$

$A \cap Z$

Palindrome $\Rightarrow x_1 = x_6 \quad x_2 = x_5 \quad x_3 = x_4$

Since A fixed $x_1 = x_2 = x_4 = T$

we get $T, T, ?, T, ?, ?$

$$x_1 = x_6 \quad x_2 = x_5, \quad x_3 = x_4$$

$\Rightarrow T, T, T, T, T, T$. All are T .

$$\therefore A \cap Z = 1.$$

$$\therefore P(A \cap Z) = \frac{1}{64}$$

$$\text{Independence test} = P(A)P(Z) = P(A \cap Z)$$

$$P(A) = \frac{1}{8} \quad \text{~~P(B)~~}$$

$$P(Z) = \frac{1}{8}$$

$$\therefore \text{Independence test} \Rightarrow \frac{1}{8} \times \frac{1}{8} = \frac{1}{64} \Rightarrow P(A \cap Z)$$

$\therefore A$ and Z are ~~not~~ independent.

b) A^c is independent of Z .

for any two events if A is independent of Z , then A^c is also independent of Z .

2.

Proof:

$$\begin{aligned} P(A^c \cap Z) &= P(Z) - P(A \cap Z) = P(Z) - P(A)P(Z) \\ &= P(Z)(1 - P(A)) \\ &= P(Z)P(A^c) \end{aligned}$$

$\therefore A^c$ is independent of Z

c) Find $P(B|Q)$

Given: $P(Q|B) = \frac{1}{4}$ $P(Q|B') = \frac{2}{7}$

$$P(B) = \left(\frac{1}{2}\right)^3 = \frac{1}{8} \quad P(B') = \frac{7}{8}$$

$$P(B|Q) = \frac{P(Q|B)P(B)}{P(Q)}$$

~~$$P(Q) = P(Q|B)P(B) + P(Q|B')$$~~

$$P(Q) = P(Q|B)P(B) + P(Q|B')P(B')$$

$$= \frac{1}{4} \times \frac{1}{8} + \frac{2}{7} \times \frac{7}{8} = \frac{1}{32} + \frac{2}{8} = \frac{1}{32} + \frac{8}{32}$$

$$= \frac{9}{32}$$

$$\therefore P(B|Q) = \frac{\frac{1}{4} \times \frac{1}{8}}{\frac{9}{32}} = \frac{\frac{1}{32}}{\frac{9}{32}} = \frac{1}{9}$$

$$\therefore P(B|Q) = \frac{1}{9}$$

4)

a) wkt $\sum p(x) = 1$.

$$\sum_{x=-3}^3 K x^2 = K [(+3)^2 + (+2)^2 + (+1)^2 + 0^2 + (-1)^2 + (-2)^2 + (-3)^2]$$

$$= K (9 + 4 + 1 + 0 + 1 + 4 + 9) = 28K = 1.$$

$$K = \frac{1}{28}$$

b) $y = |x|$

y can take non negative values 0, 1, 2, 3.

y	x: $ x = y$	$P(Y=y) = P(X = \pm y)$
0	{0}	$K \cdot 0^2 = 0$
1	{-1, 1}	$2K = 2/28 = 1/14$
2	{-2, 2}	$2 \cdot 4K = 8/28 = 2/7$
3	{-3, 3}	$2 \cdot 9K = 18/28 = 9/14$

$$\text{check } 0 + \frac{1}{14} + \frac{2}{7} + \frac{9}{14} = \frac{1+4+9}{14} = \frac{14}{14} = 1.$$

Expectation of Y

$$E[Y] = \sum_Y Y P_Y(Y) = 1\left(\frac{1}{14}\right) + 2\left(\frac{2}{7}\right) + 3\left(\frac{9}{14}\right)$$
$$= \frac{1}{14} + \frac{4}{7} + \frac{27}{14} = \frac{1+8+27}{14} = \frac{36}{14}$$

$$E[Y] = \frac{18}{7}$$

$$E[Y^2] = 1^2\left(\frac{1}{14}\right) + 2^2\left(\frac{2}{7}\right) + 3^2\left(\frac{9}{14}\right)$$
$$= \frac{1}{14} + \frac{8}{7} + \frac{81}{14} = \frac{98}{14} = 7.$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = 7 - \left(\frac{18}{7}\right)^2$$
$$= \frac{343 - 324}{49} = \frac{19}{49}.$$

$$\text{Var}(Y) = \frac{19}{49}$$
